PurePy: A Pure Functional Subset of Python for Scientific Computing

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PurePy is a pure, side-effect-free subset of Python: a functional language whose programs are referentially transparent, reproducible, and easy to reason about, yet remain ordinary Python. Every well-formed PurePy program is valid Python and runs with the same behaviour, so one source serves both a standard Python interpreter and any conforming implementation. This paper specifies PurePy as a versioned standard, with a formal grammar, well-formedness rules based on a definite-assignment analysis, and an operational semantics that preserves the meaning of well-formed programs in Python. A reference checker decides membership of the subset, and a conformance test suite is exercised on CPython, PyPy, GraalPy, and Pyodide. Aimed first at researchers in programming languages and pedagogy, PurePy is intended to grow into a common language for scientific computing, supporting portable applications in modelling, data processing, analysis, and visualisation.

1 Introduction

Python is among the most widely used languages in scientific computing, data analysis, and teaching, valued for an accessible surface syntax and a large ecosystem. That reach comes with a large and permissive language. Programs mutate state freely, depend on side effects and dynamic binding, and admit many implicit coercions, so they can be hard to reason about, hard to reproduce, and hard to target with formal tools or with implementations other than the reference interpreter. A researcher in programming languages or in pedagogy who wants a well-behaved functional language faces a trade-off: such a language is easy to reason about but unfamiliar, whereas Python is familiar but unruly.

PurePy carves out a pure, side-effect-free subset of Python that is well-behaved and easy to reason about, yet remains ordinary Python. Every well-formed PurePy program is valid Python and runs under a standard interpreter with the same result, so one artifact serves two roles: a program a scientist can run today, and a common target that a research language or tool can produce and consume. Purity makes programs referentially transparent, deterministic, and reproducible, the properties that scientific computing and formal reasoning both want.

This paper defines PurePy as a versioned formal standard: a formal grammar, well-formedness rules based on a definite-assignment analysis, and an operational semantics that preserves the meaning of well-formed programs in Python. A reference checker decides membership of the subset, and a conformance test suite exercises the standard against CPython, PyPy, GraalPy, and Pyodide. The guarantee is precise: membership is decidable, and a well-formed PurePy program is valid Python whose PurePy meaning is its Python meaning.

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The design is deliberately conservative. Rather than reconstruct Python's full behaviour, PurePy chooses a simple, well-behaved core and excludes the idioms that would break it, among them mutable state, effectful control, implicit coercions, and constructs whose meaning depends on module load order (§2). The standard is versioned, so the subset can grow as features earn their place without disturbing programs written against an earlier version.

PurePy is aimed first at researchers in programming languages and in pedagogy, and is intended to grow into a common language for scientific computing, supporting portable applications in modelling, data processing, analysis, and visualisation. Because every PurePy program is also a Python program, adopting it costs nothing at the surface, and a research language that targets PurePy inherits a large and familiar runtime.

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1.1 Contributions

After setting out the problem PurePy addresses in §2, we make the following contributions:

- a definition of PurePy as a versioned standard, with a formal grammar and well-formedness rules based on a definite-assignment analysis (§3) and an operational semantics that preserves the meaning of well-formed programs in Python (§4);
- a reference checker that decides membership of the subset, validated by a conformance test suite exercised on CPython, PyPy, GraalPy, and Pyodide (§5);
- conservative extensions to the core language, including partial application and piecewise function definitions (§6);
- case studies that use PurePy as a common functional language (§7);
- an experimental evaluation of how well AI agents generate PurePy and translate existing Python into it (§8).

§9 summarises related efforts, and §10 presents our conclusions and discusses future work.

2 Overview

PurePy is a subset of Python chosen so that its programs are well-behaved and broadly *pure*, yet run unchanged under a standard Python interpreter and compute the same result. Purity here is informal: the functional core has no observable side effects, and a name's meaning does not depend on evaluation order or on mutation. A variable, once definitely assigned, denotes a fixed value; a function is fixed by its definition; and a module's members are fixed once it has loaded. Python permits much that breaks this discipline, so PurePy tightens or excludes the idioms responsible.

The remainder of this section sets out the excluded features, grouped by the discipline each one protects, with the rationale for its exclusion. Figure 1 then summarises the Python runtime errors that well-formedness rules out.

2.1 Mutable state

A variable in PurePy denotes a fixed value once definitely assigned, so features that mutate or rebind state are excluded: `global` and `nonlocal`, which reassign bindings in an outer scope; `del`, which removes a binding; augmented assignment (`+=`, `-=` , and so on); and the walrus operator (`:=`), which assigns within an expression. Reassignment of a variable already captured by a closure is excluded for the same reason. In Python a closure captures a mutable binding, so a later reassignment is visible through the closure:

```
def f():
    x = 5
    def g():
        return x
    x = 6
    return x + g()
```

Here `g()` returns `6` in Python and the result is `12`; under a pure reading in which `x = 6` shadows the earlier binding, `g()` would return `5` and the result `11`. PurePy rejects the program, keeping closures over values rather than mutable cells, as with Java's requirement that captured locals be effectively final [Gosling et al. 2025, §15.27.2]. The rules that enforce this are given in §3.4.4.

2.2 Control flow and effects

PurePy excludes loops with `break` and `continue`, exceptions (`try`/`except`/`raise`), context managers (`with`), generators (`yield`/`yield from`), and coroutines (`async`/`await`). Functional programming has rich accounts of effects, from monads to algebraic effect handlers, so these features are not inherently at odds with a functional language. In idiomatic Python, though, they are used mainly in conjunction with mutable state, which PurePy already excludes, and iteration is expressed by comprehensions and recursion rather than by loops. A future release may add structured effects; for now the core stays effect-free.

2.3 Implicit coercions

Python accepts several implicit coercions and lax matches that are convenient but a recognised source of error, since what Python accepts rarely coincides with what the author meant. PurePy excludes three. The first is truthiness: the test of an `if` or conditional expression, or an `assert`, is coerced to `bool`, whereas PurePy requires it to be boolean already. The second is between sequence kinds in pattern matching: Python lets a list pattern match a tuple value, and a tuple pattern match a list, whereas in PurePy a list pattern matches only a list and a tuple pattern only a tuple. The third is in constructor patterns: Python lets a positional class pattern supply fewer sub-patterns than the class's field arity, matching only those given, whereas PurePy requires the exact arity.

2.4 Load-dependent meaning

A name's meaning in PurePy must not depend on the order in which modules load or on runtime state, so several of Python's import and module behaviours are excluded. Imports must appear at the start of a module, never inside a function or other block, nor after a non-import statement. Wildcard

imports (`from q import *`) are excluded, since the names they introduce are not explicit; PEP 8 recommends against them.[1] A submodule may not import itself through its parent (`from a import b` inside module `a.b`): PurePy treats the from-imported submodule as a dependency, so this is an import cycle. An `import` (as opposed to a from-import) may not name a proper descendant of the importing module (`import x.y` inside module `x`), since the binding for the head would denote the still-loading module.

A module may not bind a name that is also the name of one of its submodules. In Python an attribute of a package can be set both by an assignment in the package body and by loading the same-named submodule, and which wins depends on load order. Suppose the body of `pkg` is the single assignment `sub = 1`, and `pkg` also has a submodule `pkg.sub`:

```
import pkg                          import pkg.sub
print(pkg.sub)  # 1                 print(pkg.sub)  # <module 'pkg.sub'>
```

On the left, `pkg.sub` is never loaded, so the assignment is visible; on the right, the load overwrites the assignment. The name `pkg.sub` has no load-order-independent meaning, so PurePy prohibits the clash outright.

For the same reason, a module may not refer to a submodule it does not itself import. A submodule is not an attribute of its package until it is loaded, so `import a` followed by `a.b.f()` is rejected unless the module also imports `a.b`; Python permits the reference whenever some earlier import, anywhere in the program, happens to have loaded `a.b`. PurePy likewise rejects a reference to a module member the module body never assigns, such as `lib.x` where `lib` assigns `x` only on a branch that is not taken. In Python both cases raise `AttributeError`; PurePy rejects them statically.

2.5 Modules and classes are not first-class

For now, modules and classes are not values in PurePy. In Python a module is a value and a class declaration is an assignment, so both flow through programs as data. Treating them as first-class would complicate the static account, so the present subset keeps them second-class rather than ruling the feature out; a later release may revisit it. A module name may be projected by member access but not used as a value, and a class name appears only at constructor and pattern positions, with declarations confined to module level. Because a class is identified by its qualified name, a dataclass declaration may not re-declare a name that already denotes a class in the module; Python permits the re-declaration, leaving the earlier class reachable only through its existing instances.

2.6 Other restrictions

A few remaining restrictions do not fit the groups above. Only dataclasses are supported, so `class` definitions of other kinds are excluded.

Complementing the exclusions above, a well-formed PurePy program avoids the Python runtime errors listed in Figure 1, under the conditions noted there. Errors from ill-typed operations, such as wrong operand types, calling a non-function, or attribute access on an object lacking the attribute, are outside the scope of well-formedness.

3 Syntax

The syntax of PurePy is given in Figure 2 and Figure 3. A *block* is a non-empty sequence of statements; either a top-level statement, a `def` body, or a branch of a conditional. A *mutual region* $d_1 \dots d_{n+1}$ is a maximal contiguous non-empty sequence of `def` statements in a block: any non-`def` statement in that block, by definition, separates mutual regions. Maximality is not enforced by the

[1] https://peps.python.org/pep-0008/

Python error	Ruled out when	See
`NameError`, `UnboundLocalError`	Always: a variable may be referenced only where it is definitely assigned.	Figure 7, Figure 8
`ModuleNotFoundError`	Always: an import requires its target to be a module of the program.	Figure 10
`ImportError`	Always: a from-import requires each imported name to be a member of the source module, and import cycles are rejected. (A Python cycle may run or fail, depending on where the cycle is entered.)	Figure 9, Figure 11
`AttributeError`	Always: a module attribute reference must resolve to a definitely assigned member.	Figure 8
`TypeError`	The operation is a constructor call or class pattern, which must match the class's arity exactly. (Only excess sub-patterns are a Python error; a pattern with fewer is prohibited even though Python permits it.)	Figure 8, Figure 6

Fig. 1. Python errors prevented by well-formedness

grammar in Figure 2; the parser is responsible for grouping consecutive `def`s into a single region. The well-formedness rules treat each region as one mutually-recursive binding (see §3.4.5).

3.1 Contexts

Metavariables x, f, c range over *variables*; idiomatically f is used when the variable is expected to denote a function and c when it is expected to denote a class.

Definition 1 (Context). *A context* Γ *(or* Δ*) is a finite map from variables to one of the following* context entries θ*:*

- *the variable's* definite assignment *status* $a \in \mathbb{B} = \{\mathsf{tt}, \mathsf{ff}\}$*;*
- *a* module reference*, for a fully-qualified module name* q*: either a* stub $\mathrm{Mod}(q)$*, or* loaded *reference* $\mathrm{Mod}(q, \Gamma)$*, which also carries the member context* Γ *of* q*;*
- *a* class entry C*, of the form* $\mathrm{Cls}_\Gamma(q.c, \vec{x}, c')$*, recording the declaring context* Γ*, the qualified name* $q.c$ *of the class, sequence of distinct field names* $\vec{x}$*, and base class name* c' *(or* $\bot$ *for no base).*

Module references arise from import statements. A module declares each class name at most once, so the qualified name of a class entry is the identity of the class. The definite-assignment terminology is borrowed from Java [Gosling et al. 2025, §16]; tt indicates *definitely assigned* and ff *not definitely assigned* (or, equivalently, degenerate types "any" and "none"). It is useful to represent contexts as total functions on the set of all variables by extending the codomain with $\bot$ to indicate *undefined*. We write $\Gamma, x{:}\,\theta$ for the context that sends x to θ and otherwise agrees with Γ, define $\mathrm{dom}(\Gamma) = \{x \mid \Gamma(x) \neq \bot\}$, and write $\varnothing$ for the context where all variables are undefined. For a set of variables X, write $\mathsf{tt}(X)$ for the context $\{x{:}\,\mathsf{tt} \mid x \in X\}$; similarly $\mathsf{ff}(X)$.

Context operations. Contexts admit a non-commutative monoid of sequential composition $\Gamma \cdot \Delta$ ("override") and an idempotent commutative semigroup of parallel composition $\oplus$ ("merge"). Together these form a *right near-semiring* [van Hoorn and van Rootselaar 1967]: $\cdot$ right-distributes over $\oplus$ (Lemma 1), but left distributivity does not hold. Both operations are defined pointwise.

Definition 2 (Context override).

$$\bot \cdot \theta = \theta \cdot \bot = \theta$$
$$\theta \cdot \theta' = \theta' \quad (\textit{otherwise})$$

Override is thus right-biased and has $\bot$ as both left and right unit.

Definition 3 (Context merge). *For definite-assignment statuses $a, a' \in \mathbb{B}$:*

$$\begin{aligned} \bot \oplus \bot &= \bot \\ a \oplus \bot = \bot \oplus a &= \mathsf{ff} \\ a \oplus a' &= a \wedge a' \end{aligned}$$

Merge is otherwise undefined. Restricted to $\mathbb{B}$, it returns ff when exactly one argument is defined and the conjunction otherwise; so $\bot$ is not a unit.

Definition 4 (Context extend).

$$\begin{aligned} \mathrm{Mod}(q, \Gamma) \triangleleft \mathrm{Mod}(q, \Gamma') &= \mathrm{Mod}(q, \Gamma \triangleleft \Gamma') \\ \mathrm{Mod}(q) \triangleleft \mathrm{Mod}(q, \Gamma) = \mathrm{Mod}(q, \Gamma) \triangleleft \mathrm{Mod}(q) &= \mathrm{Mod}(q, \Gamma) \\ \bot \triangleleft \theta = \theta \triangleleft \bot &= \theta \\ \theta \triangleleft \theta' &= \theta' \quad (\textit{otherwise}) \end{aligned}$$

Extend combines module references for the same module recursively, preferring loaded references to stubs, and is otherwise right-biased like override. It is used when processing import prefixes, where two imports rooted at the same package must combine rather than shadow.

Lemma 1 (Right distributivity).

$$(\Delta \oplus \Delta') \cdot \Gamma = (\Delta \cdot \Gamma) \oplus (\Delta' \cdot \Gamma).$$

Left distributivity ($\Gamma \cdot (\Delta \oplus \Delta') = (\Gamma \cdot \Delta) \oplus (\Gamma \cdot \Delta')$) does not hold. If it did, the following two programs would be equivalent, since sequential composition with a preceding assignment could be distributed into the branches of a conditional. But they are not: the left program is ill-formed, because a variable assigned before a conditional does not remain definitely assigned if only some branches assign it (see also §3.4.2).

```
def f(b):
    x = 1
    if b:
        x = 2
    else:
        pass
    return x  # error
```

```
def f(b):
    if b:
        x = 1
        x = 2
    else:
        x = 1
        pass
    return x  # ok
```

Operators. The metavariable $\odot$ ranges over the supported binary operators and $\ominus$ over the supported unary operators, both enumerated in Figure 4. The grammar of Figure 3 is intentionally ambiguous on operator forms; disambiguation is by the precedence and associativity given in Figure 4, which follow Python.

3.2 Modules and programs

A PurePy *program* is a finite map $\mathcal{M}$ from *fully-qualified module names* (Figure 2) to *module bodies* t, of which one is a distinguished *main module* named `__main__`; here name(q) is the string literal naming q, so name(`__main__`) = `"__main__"`. We identify a module with its fully-qualified name, writing "module q" for the module named q. A module body is a sequence of imports $\vec{\imath}$ followed by a sequence of statements $\vec{s}$; since imports are not statements, they can appear neither

$$
\begin{array}{lcll}
s & ::= & \texttt{pass} & \text{(Pass)} \\
 & | & x = e & \text{(Assign)} \\
 & | & e & \text{(ExprStmt)} \\
 & | & \texttt{assert } e & \text{(Assert)} \\
 & | & \texttt{assert } e, e' & \text{(AssertMsg)} \\
 & | & \texttt{return} & \text{(ReturnNone)} \\
 & | & \texttt{return } e & \text{(Return)} \\
 & | & \texttt{if } e : \ b \ \texttt{elif } e_1 : \ b_1 \ \ldots \ \texttt{elif } e_n : \ b_n & \text{(If)} \\
 & | & \texttt{if } e : \ b \ \texttt{elif } e_1 : \ b_1 \ \ldots \ \texttt{elif } e_n : \ b_n \ \texttt{else} : \ b' & \text{(IfElse)} \\
 & | & \texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1} & \text{(Match)} \\
 & | & d_1 \ \ldots \ d_{n+1} & \text{(MutualRegion)} \\
 & | & \texttt{@dataclass class } c : \ \phi & \text{(Dataclass)} \\
 & | & \texttt{@dataclass class } c(c') : \ \phi & \text{(DataclassExtend)} \\
 & & & \\
\iota & ::= & \texttt{import } q & \text{(Import)} \\
 & | & \texttt{from } q \texttt{ import } x_1, \ldots, x_{n+1} & \text{(FromImport)} \\
 & & & \\
\phi & ::= & x_1 \texttt{ : Any } \ldots \ x_{n+1} \texttt{ : Any} & \text{(FieldList)} \\
 & | & \texttt{pass} & \text{(EmptyFieldList)} \\
 & & & \\
b & ::= & s_1 \ \ldots \ s_{n+1} & \text{(Block)} \\
 & & & \\
t & ::= & \iota_1 \ \ldots \ \iota_m \ \ s_1 \ \ldots \ s_n & \text{(ModuleBody)} \\
 & & & \\
q & ::= & x_1.x_2.\cdots.x_{n+1} & \text{(Name)} \\
 & & & \\
d & ::= & \texttt{def } f(x_1, \ldots, x_n) : \ b & \text{(Def)}
\end{array}
$$

Fig. 2. Modules and blocks

in blocks (and hence not in function bodies) nor after the first non-import statement. The well-formedness and evaluation rules check and evaluate the statements of each body $\mathcal{M}(q)$ with the assignment `__name__` = name(q) prepended, modelling the `__name__` binding the runtime supplies automatically. $\mathcal{M}$ additionally always includes the predefined modules of Figure 5, which have no body and whose contexts and environments are given directly (Definition 5), and is prefix-closed: the parent of any module in $\mathrm{dom}(\mathcal{M})$ is itself in $\mathrm{dom}(\mathcal{M})$.[2] The map $\mathcal{M}$ is fixed and ambient in all definitions that follow, as is the name q of the enclosing module in the rules that check or evaluate a single module's body.

Definition 5 (Predefined contexts and environments). *For each predefined module q of Figure 5, the context Γ_q binds the members listed for q, together with `__name__`, all definitely assigned. The dynamic semantics is parameterised by an environment ρ_q for each such q, binding the same names to implementation-defined values.*

The metafunction assigns extends by union from individual statements to blocks; on a module body $\vec{\iota}\,\vec{s}$ it acts on $\vec{s}$. Write $p \preceq q$ (*prefix*) when $q = p$ or $q = p.q'$ for some q', and $\prec$ for the strict order; the proper prefixes of a module name are its *ancestors*. The submodules metafunction is defined, with the class-related metafunctions, in Appendix A.3.

[2] A source tree encodes such an $\mathcal{M}$: each module's body is its `.py` file, each package's body is its `__init__.py`, and a package directory without an `__init__.py` encodes the empty body (a namespace package, PEP 420, https://peps.python.org/pep-0420/). Prefix-closure thus holds by construction.

e	::=	x	(Var)
	|	ℓ	(Literal)
	|	$e(e_1, \ldots, e_n)$	(Call)
	|	$q(e_1, \ldots, e_n, x_{n+1}{=}e_{n+1}, \ldots, x_{n+m}{=}e_{n+m})$	(Constr)
	|	$\{x_1{:}e_1, \ldots, x_n{:}e_n\}$	(Dict)
	|	$e.x$	(Attribute)
	|	$e[e']$	(Subscript)
	|	$e \odot e'$	(BinOp)
	|	$\ominus e$	(UnaryOp)
	|	e_1 `if` e `else` e_2	(CondExpr)
	|	$[e_1, \ldots, e_n]$	(List)
	|	$(e_1, \ldots, e_n)$	(Tuple)
	|	$[\, e\ g_1\ \ldots\ g_n]$	(ListComp)
	|	`lambda` $x_1, \ldots, x_n$: e	(Lambda)
g	::=	`if` e	(Guard)
	|	`for` x `in` e	(Generator)
ℓ	::=	n	(Number)
	|	`True`	(True)
	|	`False`	(False)
	|	`None`	(None)
p	::=	ℓ	(PatLiteral)
	|	$-n$	(PatNegLit)
	|	x	(PatVar)
	|	_	(PatWildcard)
	|	$[p_1, \ldots, p_n]$	(PatList)
	|	$(p_1, \ldots, p_n)$	(PatTuple)
	|	$q(p_1, \ldots, p_n, x_{n+1}{=}p_{n+1}, \ldots, x_{n+m}{=}p_{n+m})$	(PatConstr)
	|	p `as` x	(PatAs)

Fig. 3. Expressions and patterns

Level	Operators	Arity	Associativity
1 (lowest)	`or`	binary	left
2	`and`	binary	left
3	`not`	unary	—
4	`==` `!=` `<` `<=` `>` `>=`	binary	left
5	`+` `-`	binary	left
6	`*` `/` `//` `%`	binary	left
7	`+` `-`	unary	—
8 (highest)	`**`	binary	right

Fig. 4. Operator precedence (lowest binds loosest); follows Python

In this section, loading a module means checking its body and computing its member context (Figure 11), the static counterpart of run-time loading (§4). Both import forms load their target, together with its ancestors (Figure 10). A plain import also records each of these modules as a member of its parent (the loads-as judgement), so that the binding created for the root of the target gives attribute access along the whole chain. A from-import also loads any imported name that

Module	Members
builtins	print, len, range
math	pi, e, sqrt, exp, log, sin, cos, tan, floor, ceil
sys	argv, exit
typing	Any
dataclasses	dataclass

Fig. 5. Predefined modules and their members

denotes a submodule. An ancestor of the target is exempt from loading first if it is the importing module or one of its ancestors, since at run time it may still be loading. The exemption applies only to from-imports, because for a plain import the only ancestor that could never load first is the importing module itself, and an import naming a proper descendant of the importing module is prohibited outright (§2.4). Loading a module may thus require other modules to load first; the rules are interpreted inductively, so a program with an import cycle is ill-formed, because a module in a cycle could finish loading only after itself. Program well-formedness (Figure 11) requires the main module to be well-formed and hence, recursively, every module the program can load; a module that is never imported is not checked.

3.3 Well-formedness

3.3.1 Auxiliary definitions. Three syntactic analyses support the well-formedness rules: assigns(s), the set of variables assigned anywhere in s (without descending into nested function definitions, which have their own scope, so that assigns(s) = dom(Δ) when s : Assigns Δ); binds(p), the set of variables introduced by a pattern p; and captures(s), the set of variables from the enclosing scope that are captured by closures (lambdas or nested function definitions) within s, with companion $\text{captures}_{\text{e}}(e)$ on expressions. All are defined, by their principal cases, in Appendix A.1. captures extends to blocks and module bodies by union (like assigns), with the singleton block identified with its statement.

Definition 6 (Pattern subsumption). *The preorder $p \sqsubseteq p'$ (Appendix A.2) holds when every value matched by p is also matched by p'.*

Subsumption is used by the well-formedness rule for `match` to rule out unreachable cases and to check exhaustiveness; with the current pattern set, the latter reduces to requiring a catch-all (var or wildcard) case.

3.3.2 Well-formedness rules. A pattern list is well-formed (Figure 6) when each pattern is linear (no repeated variable names) and no earlier pattern subsumes a later one. The `match` well-formedness rules use this judgement to enforce reachability and linearity of case patterns.

Every well-formed statement either *definitely returns*, or *definitely assigns* a context Δ, which may be empty. The base cases are that `return e` definitely returns, and `x = e` definitely assigns $\{\texttt{x}\colon \texttt{tt}\}$. *Result types* R classify these outcomes: R is either Returns (definitely returns) or Assigns Δ (definitely assigns Δ). Block well-formedness is given in Figure 7; expression well-formedness in Figure 8.

Context operations $\cdot$ and $\oplus$ lift to a right semiring over result types R, with Assigns $\varnothing$ as left and right unit for $\cdot$ and Returns as unit for $\oplus$ and zero for $\cdot$.

$\Gamma \vdash p_1 \cdot \ldots \cdot p_n$

pat-lit $\dfrac{}{\Gamma \vdash \ell}$ pat-neg-lit $\dfrac{}{\Gamma \vdash -n}$ pat-var $\dfrac{}{\Gamma \vdash x}$ pat-wild $\dfrac{}{\Gamma \vdash _}$ pat-list $\dfrac{\Gamma \vdash p \quad (\forall p \in \vec{p})}{\Gamma \vdash [\vec{p}]}$ pat-tuple $\dfrac{\Gamma \vdash p \quad (\forall p \in \vec{p})}{\Gamma \vdash (\vec{p})}$

pat-as $\dfrac{\Gamma \vdash p}{\Gamma \vdash p \text{ as } x}$ pat-constr $\dfrac{\Gamma \vdash q \text{ names } C \qquad \text{fields}(C) = x_1, \ldots, x_n \cdot \vec{x}\,' \qquad \{\vec{y}\} = \{\vec{x}\,'\} \qquad \Gamma \vdash p_i \quad (\forall i)}{\Gamma \vdash q(p_1, \ldots, p_n, y_1{=}p_{n+1}, \ldots, y_m{=}p_{n+m})}$

cases-nil $\dfrac{}{\Gamma \vdash \varepsilon}$ cases-cons $\dfrac{\Gamma \vdash p \qquad \Gamma \vdash \vec{p}\,' \qquad \text{variables in } p \text{ distinct} \qquad p' \not\sqsubseteq p \quad (\forall p' \in \vec{p}\,')}{\Gamma \vdash p \cdot \vec{p}\,'}$

Fig. 6. Case patterns are well-formed under Γ

Definition 7 (Override and merge on result types).

$$\text{Assigns } \Delta \oplus \text{Assigns } \Delta' = \text{Assigns } (\Delta \oplus \Delta')$$
$$R \oplus \text{Returns} = \text{Returns} \oplus R = R$$

$$\text{Assigns } \Delta \cdot \text{Assigns } \Delta' = \text{Assigns } (\Delta \cdot \Delta')$$
$$R \cdot \text{Returns} = \text{Returns} \cdot R = \text{Returns}$$

The $\text{Returns} \cdot R$ case never arises because the well-formedness judgement prohibits unreachable statements (§3.4.7).

Lemma 2 (Right distributivity (result types)).

$$(R \oplus R') \cdot S = (R \cdot S) \oplus (R' \cdot S)$$

A module attribute reference is well-formed only when it resolves to a definitely assigned member (rule attr-module); see §2.4.

The output context Γ of the module well-formedness judgement (Figure 11) comprises the classes and variables a module defines, together with its intrinsic `__name__` and a stub for each submodule, never the names it imports: following PEP 484[3], an imported name is a private dependency, not re-exported as part of the interface. Submodule stubs and the module's own bindings are disjoint, because the module rules prohibit a module from binding a name that is also the name of a submodule (§2.4).

3.4 Examples

3.4.1 Assignment as definition. Although PurePy is pure and all variable usage is well-scoped, local variables are introduced using an *assignment statement*, for compatibility with Python. If a variable is definitely assigned in any branch of a conditional, then for that variable to be usable after the conditional, every other branch of the conditional must either definitely return, or definitely assign that variable too. Thus the following are both well-formed:

[3] https://peps.python.org/pep-0484/

$\Gamma \vdash b : R$

pass

$$\frac{}{\Gamma \vdash \texttt{pass} : \text{Assigns } \varnothing}$$

assign

$$\frac{\Gamma \vdash e \qquad x \notin \text{captures}_e(e)}{\Gamma \vdash x = e : \text{Assigns } \{x : \text{tt}\}}$$

expr-stmt

$$\frac{\Gamma \vdash e}{\Gamma \vdash e : \text{Assigns } \varnothing}$$

assert

$$\frac{\Gamma \vdash e}{\Gamma \vdash \texttt{assert } e : \text{Assigns } \varnothing}$$

assert-msg

$$\frac{\Gamma \vdash e \qquad \Gamma \vdash e'}{\Gamma \vdash \texttt{assert } e, e' : \text{Assigns } \varnothing}$$

return-none

$$\frac{}{\Gamma \vdash \texttt{return} : \text{Returns}}$$

return

$$\frac{\Gamma \vdash e}{\Gamma \vdash \texttt{return } e : \text{Returns}}$$

if

$$\frac{\Gamma \vdash e_i \quad (\forall i) \qquad \Gamma \vdash b_i : R_i \quad (\forall i)}{\Gamma \vdash \texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1} : R_1 \oplus \ldots R_{n+1} \oplus \ldots \oplus \text{Assigns } \varnothing}$$

if-else

$$\frac{\Gamma \vdash e_i \quad (\forall i) \qquad \Gamma \vdash b_i : R_i \quad (\forall i)}{\Gamma \vdash \texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1} \ \texttt{else} : \ b_{n+2} : R_1 \oplus \ldots \oplus R_{n+2}}$$

def

$$\frac{\Gamma \cdot \text{tt}(\vec{f} \cdot \vec{x}_i) \cdot \text{ff}(\text{assigns}(b_i) \setminus \vec{x}_i) \vdash b_i : R_i \quad (\forall i) \qquad \vec{f} \text{ distinct}}{\Gamma \vdash \texttt{def } f_1(\vec{x}_1) : \ b_1 \cdot \ldots \cdot \texttt{def } f_{n+1}(\vec{x}_{n+1}) : \ b_{n+1} : \text{Assigns } \text{tt}(\vec{f})}$$

match

$$\frac{\Gamma \vdash e \qquad \Gamma \cdot \text{tt}(\text{binds}(p_i)) \vdash b_i : R_i \quad (\forall i) \qquad S_i = \text{Assigns } \text{tt}(\text{binds}(p_i)) \cdot R_i \quad (\forall i) \qquad \Gamma \vdash \vec{p} \qquad p_{n+1} \in \{x, _\}}{\Gamma \vdash \texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1} : S_1 \oplus \ldots \oplus S_{n+1}}$$

match-partial

$$\frac{\Gamma \vdash e \qquad \Gamma \cdot \text{tt}(\text{binds}(p_i)) \vdash b_i : R_i \quad (\forall i) \qquad S_i = \text{Assigns } \text{tt}(\text{binds}(p_i)) \cdot R_i \quad (\forall i) \qquad \Gamma \vdash \vec{p} \qquad p_{n+1} \notin \{x, _\}}{\Gamma \vdash \texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1} : S_1 \oplus \ldots \oplus S_{n+1} \oplus \text{Assigns } \varnothing}$$

cons

$$\frac{\Gamma \vdash s : \text{Assigns } \Delta \qquad \Gamma \cdot \Delta \vdash b : R \qquad \text{captures}(s) \cap \text{assigns}(b) = \varnothing}{\Gamma \vdash s \cdot b : \text{Assigns } \Delta \cdot R}$$

Fig. 7. Block b is well-formed in Γ

```
if a < b:
    comp = "smaller"
else:
    comp = "bigger"
return comp
```

```
if a < b:
    return "smaller"
else:
    comp = "bigger"
return comp
```

The definite assignment set for a conditional that has at least one branch which definitely assigns is calculated by merging the definite assignment sets for all branches which definitely assign, using the $\oplus$ operator on environments.

$\Gamma \vdash q$ names θ

simple-var
$$\frac{x{:}\,\mathrm{tt} \in \Gamma}{\Gamma \vdash x \text{ names } \mathrm{tt}}$$

simple-module
$$\frac{x{:}\,\mathrm{Mod}(q, \Gamma') \in \Gamma}{\Gamma \vdash x \text{ names } \mathrm{Mod}(q, \Gamma')}$$

simple-class
$$\frac{c{:}\,C \in \Gamma}{\Gamma \vdash c \text{ names } C}$$

qualified
$$\frac{\Gamma \vdash q \text{ names } \mathrm{Mod}(q', \Gamma') \qquad \Gamma' \vdash x \text{ names } \theta}{\Gamma \vdash q.x \text{ names } \theta}$$

(a) Name q resolves to context entry θ under Γ

$\Gamma \vdash e$

var
$$\frac{x{:}\,\mathrm{tt} \in \Gamma}{\Gamma \vdash x}$$

var-builtin
$$\frac{x \notin \mathrm{dom}(\Gamma) \qquad x{:}\,\mathrm{tt} \in \Gamma_{\texttt{builtins}}}{\Gamma \vdash x}$$

lambda
$$\frac{\Gamma \cdot \mathrm{tt}(\vec{x}) \vdash e}{\Gamma \vdash \texttt{lambda}\ \vec{x}{:}\ e}$$

constr
$$\frac{\Gamma \vdash q \text{ names } C \qquad \mathrm{fields}(C) = x_1, \ldots, x_n \cdot \vec{x}\,' \qquad \{\vec{y}\} = \{\vec{x}\,'\} \qquad \Gamma \vdash e_i \quad (\forall i)}{\Gamma \vdash q(e_1, \ldots, e_n, y_1{=}e_{n+1}, \ldots, y_m{=}e_{n+m})}$$

attr-module
$$\frac{\Gamma \vdash e.x \text{ names } \mathrm{tt}}{\Gamma \vdash e.x}$$

attr-object
$$\frac{\Gamma \vdash e}{\Gamma \vdash e.x}$$

(b) Expression e is well-formed under Γ

Fig. 8. Expression well-formedness

Γ, x imports θ

imp-member
$$\frac{\Gamma \vdash x \text{ names } \theta}{\Gamma, x \text{ imports } \theta}$$

imp-submodule
$$\frac{x{:}\,\mathrm{Mod}(q) \in \Gamma \qquad \vdash_{\mathsf{load}} q : \Delta}{\Gamma, x \text{ imports } \mathrm{Mod}(q, \Delta)}$$

Fig. 9. Name x imported from module context Γ has entry θ

Definite assignment thus gives each reference a unique lexical referent: a variable is introduced by an assignment, possibly jointly by the branches of a conditional, and a name may be referenced only where one such introduction covers every path. This is stricter than avoiding Python's `NameError`: the left-hand program of §3.1 assigns `x` on every run, but is rejected because the reference would denote the branch-local variable on one path and the variable it hides on the other.

3.4.2 Local assignment in conditional clauses. For flexibility, we permit *branch-local* variables. The assignment of `x` below (left) is permitted, and can be used locally in that branch of the conditional:

$\vdash \vec{\imath} : \Gamma$

imports-nil

$$\frac{}{\vdash \varepsilon : \varnothing}$$

import

$$\frac{q' = x \cdot \vec{x} \qquad q \not\prec q' \qquad \vdash_{\text{load}} q' : \Delta \qquad q', \text{Mod}(q', \Delta) \text{ loads-as } \theta \qquad \vdash \vec{\imath} : \Gamma}{\vdash \texttt{import}\ q' \cdot \vec{\imath} : \{x{:}\,\theta\} \triangleleft \Gamma}$$

from-import

$$\frac{\vdash_{\text{load}} q' : \Delta \qquad \vdash_{\text{load}} p : \Delta_p \quad (\forall p \prec q', \ p \not\preceq q) \qquad \Delta, x_i \text{ imports } \theta_i \quad (\forall i) \qquad \vdash \vec{\imath} : \Gamma}{\vdash \texttt{from}\ q'\ \texttt{import}\ x_1, \ldots, x_{n+1} \cdot \vec{\imath} : \{x_1{:}\,\theta_1, \ldots, x_{n+1}{:}\,\theta_{n+1}\} \triangleleft \Gamma}$$

(a) Import prefix $\vec{\imath}$ of module q produces bindings Γ

q, θ loads-as θ'

loads-as-simple

$$\frac{}{x, \theta \text{ loads-as } \theta}$$

loads-as-qualified

$$\frac{\vdash_{\text{load}} q : \Gamma' \qquad q, \text{Mod}(q, \Gamma' \triangleleft \{x{:}\,\theta\}) \text{ loads-as } \theta'}{q.x, \theta \text{ loads-as } \theta'}$$

(b) Reference θ for module q loads as the reference θ' for the root of q, loading the ancestors of q

Fig. 10. Import well-formedness

```
def foo():
  z = 6
  if b:
    x = 3
    y = x + 1
  else:
    y = 0
  return z
```

(a) Branch-local definition of `x`

```
def foo():
  x = 6
  if b:
    x = 3
    y = x + 1
  else:
    y = 0
  return x # error: partially defined
```

(b) Now with use of `x` after the conditional

However such variables only appear to be branch-local. The definition of $\oplus$ actually ensures that a variable assigned in any branch is considered assigned by the conditional; however unless it is assigned in *every* branch that does not definitely return, it will be bound to ff in the corresponding environment. Any attempt to refer to such a variable is illegal, as it refers to a variable which may not be assigned. (Thus the right-hand program is ill-formed, despite the enclosing `x = 6`, because in the `return x` statement, the inner assignment shadows the the outer one, and the inner assignment is only partial.) If we were to allow the program on the right, there would be no straightforward way to assign it a semantics which is both pure, and in agreement with Python, since in Python the function returns `3` in the case where `b` is `True`.

3.4.3 Function-level scoping of local variables. In Python, all local variables are scoped to the entire function body, not just from the point of assignment onwards. The assigns function in the mutual rule reflects this: all variables assigned anywhere in the function body are introduced at ff before the body is checked. This means the following program is statically rejected by PurePy, whereas Python rejects it at runtime with an `UnboundLocalError`:

```
y = 7
```

$\Gamma \vdash \vec{s} : \Delta$

nil

$$\frac{}{\Gamma \vdash \varepsilon : \varnothing}$$

statement

$$\frac{\Gamma \vdash s : \text{Assigns } \Delta \qquad \Gamma \cdot \Delta \vdash \vec{s} : \Delta' \qquad \text{captures}(s) \cap \text{assigns}(\vec{s}) = \varnothing}{\Gamma \vdash s \cdot \vec{s} : \Delta \cdot \Delta'}$$

class

$$\frac{\nexists C'.\ c{:}C' \in \Gamma \qquad \text{own-fields}(\phi) = \vec{x}\ \text{distinct} \qquad C = \text{Cls}_\Gamma(q.c, \vec{x}, \bot) \qquad \Gamma, c{:}C \vdash \vec{s} : \Delta}{\Gamma \vdash \texttt{@dataclass class } c : \ \phi \cdot \vec{s} : \{c{:}C\} \cdot \Delta}$$

class-extend

$$\frac{\nexists C'.\ c{:}C' \in \Gamma \qquad c' {:} \text{Cls}_\Delta(q.c', \vec{y}, c'') \in \Gamma \qquad \text{own-fields}(\phi) = \vec{x}\ \text{distinct} \qquad \vec{x} \cap \text{fields}(\Gamma(c')) = \varnothing \qquad C = \text{Cls}_\Gamma(q.c, \vec{x}, c') \qquad \Gamma, c{:}C \vdash \vec{s} : \Delta}{\Gamma \vdash \texttt{@dataclass class } c(c') : \ \phi \cdot \vec{s} : \{c{:}C\} \cdot \Delta}$$

(a) Statements $\vec{s}$ of module q are well-formed in Γ

$\vdash_{\text{load}} q : \Gamma$

predefined

$$\frac{q \text{ predefined}}{\vdash_{\text{load}} q : \Gamma_q}$$

module

$$\frac{\mathcal{M}(q) = \vec{\imath}\,\vec{s} \qquad \vdash \vec{\imath} : \Gamma \qquad \Gamma \vdash (\texttt{__name__} = \text{name}(q)) \cdot \vec{s} : \Delta \qquad (\text{dom}(\Gamma) \cup \text{dom}(\Delta)) \cap \text{dom}(\text{submodules}(q)) = \varnothing}{\vdash_{\text{load}} q : \text{submodules}(q) \cdot \Delta}$$

$\vdash \mathcal{M}$

program

$$\frac{\vdash_{\text{load}} \texttt{__main__} : \Gamma}{\vdash \mathcal{M}}$$

(b) Module and program well-formedness

Fig. 11. Module well-formedness

```
def f():
    x = y  # error: y is local (due to assignment below)
    y = 8

f()
```

Because `y` appears on the left-hand side of an assignment in `f`, it is in assigns(b) for the body of `f`, and is therefore introduced at ff at the top of the function. The reference to `y` in `x = y` then fails the well-formedness check, since $\Gamma(\texttt{y}) = \text{ff}$ rather than $\Gamma(\texttt{y}) = \text{tt}$.

3.4.4 Reassignment of captured variables. The exclusion of captured-variable reassignment (§2.1) is enforced by a side condition on the seq rule: no variable captured by a statement s may be reassigned in the rest of the block b. In the example of §2.1, `x` is captured by the definition of `g` and

reassigned by `x = 6`, so the program is rejected. A dataclass declaration counts as an assignment of its class name for these definitions, though re-declaring a class name is prohibited outright (§2.5).

The same issue arises when a variable is captured and reassigned in the *same* statement:

```
x = 5
def x():
    return x
x()  # Python: returns the function (late binding); pure: returns 5
```

The side condition on the assign rule rejects programs like this, requiring that the variable being bound is not captured by the expression. The mutual rule, by contrast, binds each f_i in the typing context for every body b_i, so the function names may legitimately appear in captures of those bodies. This is what permits self-recursion (the singleton-region case) and mutual recursion within a region.

3.4.5 Mutual recursion. The mutual rule binds the names $f_1, \ldots, f_{n+1}$ of a mutual region $d_1 \ldots d_{n+1}$ simultaneously. Each body b_i is checked in a context that includes all the f_j as definitely assigned, so any f_j may reference any other f_k within the region (the singleton case $n = 0$ permits self-recursion). The names in a region must be distinct.

Inserting any non-`def` statement between two `def`s splits them into separate regions, breaking the mutual reference:

```
def even(n):
    if n == 0:
        return True
    return odd(n - 1)

x = even(0)  # intervening statement separates the regions

def odd(n):
    if n == 0:
        return False
    return even(n - 1)
```

This program is rejected statically: `even` forms a singleton region in which `odd` is not yet bound, so the reference to `odd` in its body fails the var rule. Python accepts it: the call `even(0)` does not actually reach `odd`, and Python's late binding means `odd` is only looked up when needed.

Each branch of a conditional may contain its own mutual regions, subject to the usual definite assignment rules. The following is well-formed because both branches define both names:

```
b = True

if b:
    def f():
        return g()
    def g():
        return "via mutual region"
else:
    def f():
        return "f independent"
    def g():
        return "g independent"
```

```
print(f())
print(g())
```

A region containing two definitions of the same name is rejected directly by the mutual rule's distinct-names side condition:

```
def g():
    return 0

def f():
    return g()
def g():     # error: g defined twice in the same region
    return 1
```

(Python accepts this; late binding makes `f`'s call to `g` resolve to the second `def g`.)

Wrapping the second `def f; def g` pair in a conditional sidesteps the duplicate-name issue: the outer `def g` and the inner `def g` are now in different blocks, so they no longer fall in the same region. One might still expect a captured-variable problem, since the inner `def g` could be seen as rebinding the outer `g` that `f` captures. No such problem arises:

```
def g():
    return 0

if b:
    def f():
        return g()
    def g():
        return 1
```

The inner `def f; def g` form a branch-local mutual region, so `f`'s body resolves `g` to its sibling rather than to the outer `g`. The two are bound simultaneously, so `f` cannot be called before the inner `g` exists, and the value `f` sees is the same as the value Python's late binding finds at call time.

3.4.6 Pattern variables escape the match. A variable bound by a `case` pattern is in scope after the `match` statement, just as a variable assigned inside a conditional branch is in scope after the conditional. Python treats matching this way; PurePy follows.

```
v = 5
match v:
    case x:
        pass
print(x)    # prints 5
```

Here the catch-all `case x:` binds `x` in the body's scope, and the binding survives into the enclosing scope. Definite assignment after the match follows the same merge logic as a conditional: if every case binds the variable (or returns), it is definitely assigned afterwards; otherwise it is not (see also §3.4.2).

3.4.7 Unreachable code. A non-empty sequence of statements returns only if the last statement returns, and all preceding statements assign. Thus the following is illegal:

```
def foo(x):
  return x + 1
```

$$
\begin{array}{rcll}
v & ::= & \ell & \text{(Literal)} \\
 & | & [v_1, \ldots, v_n] & \text{(List)} \\
 & | & (v_1, \ldots, v_n) & \text{(Tuple)} \\
 & | & \mathsf{Lam}_\Gamma(\rho, \vec{x}, e) & \text{(LambdaClosure)} \\
 & | & \mathsf{Def}_\Gamma(\rho, \vec{d}, i) & \text{(DefClosure)} \\
 & | & \mathsf{Obj}_\Gamma(C, \rho) & \text{(Object)} \\
 & | & \mathsf{Mod}(q) & \text{(Module stub)} \\
 & | & \mathsf{Mod}(q, \rho) & \text{(Loaded module)} \\
 & | & \mathsf{Cls}_\Gamma(q.c, \vec{x}, c') & \text{(Class)} \\
\rho & ::= & \{x_1 : v_1, \ldots, x_n : v_n\} & \text{(Environment)} \\
r & ::= & \mathsf{assigns}\ \rho & \\
 & | & \mathsf{returns}\ v &
\end{array}
$$

Fig. 14. Values, environments and evaluation results

```
return x    # error: unreachable
```

3.4.8 Conditionals without `else` clauses. An `if` statement without an `else` clause is equivalent to an `if` statement with an `else` clause whose body is a `pass` statement; such a statement never returns.

3.4.9 Implicit `None`. A `return` statement without an expression is equivalent to `return None`.

3.4.10 Implicit `return`. A function whose body s assigns rather than returns is equivalent to a function whose body is s followed by `return None`. This ensures every function returns, even if it is not explicitly specified.

3.4.11 `assert` without a message. An `assert e` statement without a message is equivalent to `assert e, ""`.

4 Operational semantics

Values, environments, and evaluation results are given in Figure 14. For a statement s, the result assigns ρ represents *normal completion* of s, with accumulated environment ρ; the result returns v represents *abrupt completion* of s with return value v. Composing statements sequentially composes their results according to the following partial monoid, analogous to sequential composition of result types:

$$
\begin{aligned}
\mathsf{assigns}\ \rho \cdot \mathsf{assigns}\ \rho' &= \mathsf{assigns}\ (\rho \cdot \rho') \\
\mathsf{assigns}\ \rho \cdot \mathsf{returns}\ v &= \mathsf{returns}\ v
\end{aligned}
$$

with assigns $\varnothing$ as left and right unit.

A value is well-formed at a context entry (Figure 15): ordinary values at tt, module references and class entries at the corresponding entry. An environment is well-formed at a context that binds its variables at corresponding entries.

Module and program evaluation are given in Figure 18; import evaluation in Figure 17; importing names from a loaded module in Figure 16. Extend (Definition 4) extends to module values by reading its equations with a loaded reference $\mathsf{Mod}(q, \rho)$ carrying an environment in place of a context. The loading judgement $q \Rightarrow_{\mathsf{load}} \rho$ evaluates the import prefix of q and then its statements, mirroring the well-formedness judgement $\vdash_{\mathsf{load}} q : \Gamma$ (Figure 11); as with the static rules, the mutual recursion between loading and evaluation is grounded by program well-formedness, which rules out import

$\vdash v : \theta$

$$\textsc{val-lit}\ \frac{}{\vdash \ell : \mathtt{tt}} \qquad \textsc{val-list}\ \frac{\vdash v_i : \mathtt{tt} \quad (\forall i)}{\vdash [v_1, \ldots, v_n] : \mathtt{tt}} \qquad \textsc{val-tuple}\ \frac{\vdash v_i : \mathtt{tt} \quad (\forall i)}{\vdash (v_1, \ldots, v_n) : \mathtt{tt}} \qquad \textsc{val-lambda}\ \frac{\vdash \rho : \Gamma \qquad \Gamma \cdot \mathtt{tt}(\vec{x}) \vdash e}{\vdash \mathsf{Lam}_\Gamma(\rho, \vec{x}, e) : \mathtt{tt}}$$

$$\textsc{val-def}\ \frac{\vdash \rho : \Gamma \qquad \Gamma \cdot \mathtt{tt}(\vec{f} \cdot \vec{x}_j) \cdot \mathtt{ff}(\mathsf{assigns}(b_j) \setminus \vec{x}_j) \vdash b_j : R_j \quad (\forall j) \qquad \vec{f}\ \text{distinct}}{\vdash \mathsf{Def}_\Gamma(\rho, \mathtt{def}\ f_1(\vec{x}_1) :\ b_1 \cdot \ldots \cdot \mathtt{def}\ f_{n+1}(\vec{x}_{n+1}) :\ b_{n+1}, i) : \mathtt{tt}} \qquad \textsc{val-obj}\ \frac{\vdash C : C \qquad \vdash \rho : \Gamma}{\vdash \mathsf{Obj}_\Gamma(C, \rho) : \mathtt{tt}}$$

$$\textsc{val-mod}\ \frac{q \in \mathrm{dom}(\mathcal{M})}{\vdash \mathsf{Mod}(q) : \mathsf{Mod}(q)} \qquad \textsc{val-mod-loaded}\ \frac{\vdash \rho : \Gamma}{\vdash \mathsf{Mod}(q, \rho) : \mathsf{Mod}(q, \Gamma)} \qquad \textsc{val-class}\ \frac{}{\vdash \mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot) : \mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot)}$$

$$\textsc{val-class-extend}\ \frac{c' : C' \in \Gamma}{\vdash \mathsf{Cls}_\Gamma(q.c, \vec{x}, c') : \mathsf{Cls}_\Gamma(q.c, \vec{x}, c')}$$

$\vdash \rho : \Gamma$

$$\textsc{env-nil}\ \frac{}{\vdash \varnothing : \varnothing} \qquad \textsc{env-cons}\ \frac{\vdash v : \theta \qquad \vdash \rho : \Gamma}{\vdash \{x : v\} \cdot \rho : \{x : \theta\} \cdot \Gamma}$$

Fig. 15. Well-formed values and environments

ρ, x imports v

$$\textsc{eval-imp-member}\ \frac{x : v \in \rho \qquad v \neq \mathsf{Mod}(q') \quad (\forall q')}{\rho, x \text{ imports } v} \qquad \textsc{eval-imp-submodule}\ \frac{x : \mathsf{Mod}(q') \in \rho \qquad q' \Rightarrow_{\mathsf{load}} \rho'}{\rho, x \text{ imports } \mathsf{Mod}(q', \rho')}$$

Fig. 16. Name x imported from module environment ρ has value v

cycles (§3.2). Loading takes a single module; the ancestors of an imported module are loaded by the loads-as judgement for an import, and directly as premises of eval-from-import for a from-import.

The correspondence is exact: if $\vdash_{\mathsf{load}} q : \Gamma$ and $q \Rightarrow_{\mathsf{load}} \rho$ then $\vdash \rho : \Gamma$ (Figure 15). Since imports precede statements, every module a statement can reference is loaded before the statement runs, and since loading has no observable effects, reloading a module is unobservable; the semantics therefore needs no cache of loaded modules. These rules use the environment $\rho_{\mathtt{builtins}}$ binding the names of Figure 5. A program evaluates to its exit status: a run whose main module loads successfully yields 0 (rule eval-program).

Block evaluation $\rho, b \Rightarrow r$ is given in Figure 19. Function definitions delimit the scope of sequential-composition effects, so as black boxes functions are always pure.

Expression evaluation $\rho, e \Rightarrow v$ is given in Figure 20. The pattern-matching judgement $\rho, p, v \rightsquigarrow \rho'$ (Figure 21) is a deterministic partial function; we write $\rho, p, v \not\rightsquigarrow$ (with no right-hand side) for $\nexists \rho'.\ \rho, p, v \rightsquigarrow \rho'$, i.e. matching is undefined. A constructor pattern matches any object whose class has the pattern's class among its ancestors; sub-patterns bind the fields of the pattern's class. An

$\vec{\imath} \Rightarrow \rho$

eval-imports-nil

$$\frac{}{\varepsilon \Rightarrow \varnothing}$$

eval-import

$$\frac{q' = x \cdot \vec{x} \qquad q' \Rightarrow_{\text{load}} \rho \qquad q', \text{Mod}(q', \rho) \text{ loads-as } v \qquad \vec{\imath} \Rightarrow \rho'}{\texttt{import } q' \cdot \vec{\imath} \Rightarrow \{x{:}\, v\} \lhd \rho'}$$

eval-from-import

$$\frac{q' \Rightarrow_{\text{load}} \rho \qquad p \Rightarrow_{\text{load}} \rho_p \quad (\forall p \prec q',\ p \not\equiv q) \qquad \rho, x_i \text{ imports } v_i \quad (\forall i) \qquad \vec{\imath} \Rightarrow \rho''}{\texttt{from } q' \texttt{ import } x_1, \ldots, x_{n+1} \cdot \vec{\imath} \Rightarrow \{x_1{:}\, v_1, \ldots, x_{n+1}{:}\, v_{n+1}\} \lhd \rho''}$$

(a) Import prefix $\vec{\imath}$ of module q produces bindings ρ

q, v loads-as v'

eval-loads-as-simple

$$\frac{}{x, v \text{ loads-as } v}$$

eval-loads-as-qualified

$$\frac{q \Rightarrow_{\text{load}} \rho \qquad q, \text{Mod}(q, \rho \lhd \{x{:}\, v\}) \text{ loads-as } v'}{q.x, v \text{ loads-as } v'}$$

(b) Reference v for module q loads as the reference v' for the root of q, loading the ancestors of q

Fig. 17. Import evaluation

$\rho, \vec{s} \Rightarrow \rho'$

eval-nil

$$\frac{}{\rho, \varepsilon \Rightarrow \varnothing}$$

eval-statement

$$\frac{\rho, s \Rightarrow \text{assigns } \rho' \qquad \rho \cdot \rho', \vec{s} \Rightarrow \rho''}{\rho, s \cdot \vec{s} \Rightarrow \rho' \cdot \rho''}$$

eval-class

$$\frac{C = \text{Cls}_\Gamma(q.c, \text{own-fields}(\phi), \bot) \qquad \rho \cdot \{c{:}\, C\}, \vec{s} \Rightarrow \rho'}{\rho, \texttt{@dataclass class } c:\ \phi \cdot \vec{s} \Rightarrow \{c{:}\, C\} \cdot \rho'}$$

eval-class-extend

$$\frac{C = \text{Cls}_\Gamma(q.c, \text{own-fields}(\phi), c') \qquad \rho \cdot \{c{:}\, C\}, \vec{s} \Rightarrow \rho'}{\rho, \texttt{@dataclass class } c(c'):\ \phi \cdot \vec{s} \Rightarrow \{c{:}\, C\} \cdot \rho'}$$

(a) Statements $\vec{s}$ of module q evaluate under environment $\rho : \Gamma$, producing bindings ρ'

$q \Rightarrow_{\text{load}} \rho$

eval-predefined

$$\frac{q \text{ predefined}}{q \Rightarrow_{\text{load}} \rho_q}$$

load

$$\frac{\mathcal{M}(q) = \vec{\imath}\, \vec{s} \qquad \vec{\imath} \Rightarrow \rho \qquad \rho, (\texttt{__name__} = \text{name}(q)) \cdot \vec{s} \Rightarrow \rho'}{q \Rightarrow_{\text{load}} \text{submodules}(q) \cdot \rho'}$$

$\mathcal{M} \Rightarrow n$

eval-program

$$\frac{\texttt{__main__} \Rightarrow_{\text{load}} \rho}{\mathcal{M} \Rightarrow 0}$$

(b) Module loading and program evaluation

Fig. 18. Module and program evaluation

$$\mathrm{dispatch}(\rho, v, \varepsilon) = (\varnothing, \texttt{pass})$$
$$\mathrm{dispatch}(\rho, v, (\texttt{case } p\texttt{: } b) \cdot \vec{c}) = (\rho', b) \qquad \text{if } \rho, p, v \rightsquigarrow \rho'$$
$$\mathrm{dispatch}(\rho, v, (\texttt{case } p\texttt{: } b) \cdot \vec{c}) = \mathrm{dispatch}(\rho, v, \vec{c}) \qquad \text{if } \rho, p, v \not\rightsquigarrow$$

$\rho, b \Rightarrow r$

eval-pass
$$\frac{}{\rho, \texttt{pass} \Rightarrow \text{assigns } \varnothing}$$

eval-assign
$$\frac{\rho, e \Rightarrow v}{\rho, x = e \Rightarrow \text{assigns } \{x\colon v\}}$$

eval-expr-stmt
$$\frac{\rho, e \Rightarrow v}{\rho, e \Rightarrow \text{assigns } \varnothing}$$

eval-assert
$$\frac{\rho, e \Rightarrow v \qquad v \neq \texttt{False}}{\rho, \texttt{assert } e \Rightarrow \text{assigns } \varnothing}$$

eval-assert-msg
$$\frac{\rho, e \Rightarrow v \qquad v \neq \texttt{False}}{\rho, \texttt{assert } e, e' \Rightarrow \text{assigns } \varnothing}$$

eval-return
$$\frac{\rho, e \Rightarrow v}{\rho, \texttt{return } e \Rightarrow \text{returns } v}$$

eval-def
$$\frac{d_i = \texttt{def } f_i(\vec{x}_i)\texttt{: } b_i \quad (\forall i) \qquad v_i = \mathrm{Def}_\Gamma(\rho, \vec{d}, i) \quad (\forall i)}{\rho, \vec{d} \Rightarrow \text{assigns } \{\vec{f}\colon \vec{v}\}}$$

eval-if
$$\frac{\rho, e_j \Rightarrow \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \texttt{True} \qquad \rho, b_i \Rightarrow r}{\rho, \texttt{if } e_1\texttt{: } b_1 \texttt{ elif } e_2\texttt{: } b_2 \ \ldots \ \texttt{elif } e_{n+1}\texttt{: } b_{n+1} \texttt{ else: } b_{n+2} \Rightarrow r}$$

eval-else
$$\frac{\rho, e_j \Rightarrow \texttt{False} \quad (\forall j \in \{1, \ldots, n+1\}) \qquad \rho, b_{n+2} \Rightarrow r}{\rho, \texttt{if } e_1\texttt{: } b_1 \texttt{ elif } e_2\texttt{: } b_2 \ \ldots \ \texttt{elif } e_{n+1}\texttt{: } b_{n+1} \texttt{ else: } b_{n+2} \Rightarrow r}$$

eval-match
$$\frac{\rho, e \Rightarrow v \qquad \mathrm{dispatch}(\rho, v, \texttt{case } p_1\texttt{: } b_1 \cdot \ldots \cdot \texttt{case } p_{n+1}\texttt{: } b_{n+1}) = (\rho', b) \qquad \rho \cdot \rho', b \Rightarrow r}{\rho, \texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1} \Rightarrow r}$$

eval-cons
$$\frac{\rho, s \Rightarrow \text{assigns } \rho' \qquad \rho \cdot \rho', b \Rightarrow r}{\rho, s \cdot b \Rightarrow \text{assigns } \rho' \cdot r}$$

Fig. 19. Block b evaluates to result r under environment $\rho : \Gamma$

object records its class entry, whose qualified name is the identity of the class: since a module declares each class name at most once, the entry comparison in pat-constr amounts to a comparison of qualified names.

5 Reference implementation

A reference checker, written in Python over the `ast` module, decides membership of PurePy and reports why a program is rejected. This section describes the checker and its correspondence with the well-formedness rules of §3.

5.1 Testing strategy

A conformance test suite pairs each test with the verdict Python itself gives, and runs on CPython, PyPy, GraalPy, and Pyodide. This section sets out the test taxonomy and the differential and metamorphic strategies used to build it.

$$\text{unwrap}(\text{returns } v) = v$$
$$\text{unwrap}(\text{assigns } \rho) = \texttt{None}$$

$$\text{env}(\rho, x) = \rho' \quad \text{where } \rho(x) = \text{Mod}(q', \rho')$$
$$\text{env}(\rho, q.x) = \rho' \quad \text{where } \text{env}(\rho, q)(x) = \text{Mod}(q', \rho')$$

$$\text{resolve-class}(\rho, c) = \rho(c)$$
$$\text{resolve-class}(\rho, q.c) = \text{env}(\rho, q)(c)$$

$\rho, e \Rightarrow v$

eval-var
$$\frac{x : v \in \rho}{\rho, x \Rightarrow v}$$

eval-var-builtin
$$\frac{x \notin \text{dom}(\rho) \qquad x : v \in \rho_{\texttt{builtins}}}{\rho, x \Rightarrow v}$$

eval-lambda
$$\frac{}{\rho, \texttt{lambda } x_1, \ldots, x_n : e \Rightarrow \text{Lam}_\Gamma(\rho, \vec{x}, e)}$$

eval-call-lambda
$$\frac{\rho, e \Rightarrow \text{Lam}_\Delta(\rho', \vec{x}, e') \qquad \rho, \vec{e} \Rightarrow \vec{u} \qquad \rho' \cdot \{\vec{x} : \vec{u}\}, e' \Rightarrow v}{\rho, e(e_1, \ldots, e_n) \Rightarrow v}$$

eval-call-def
$$\frac{\rho, e \Rightarrow \text{Def}_\Delta(\rho', \vec{d}, j) \qquad d_i = \texttt{def } f_i(\vec{x}_i) : \ b_i \quad (\forall i) \qquad v_i = \text{Def}_\Delta(\rho', \vec{d}, i) \quad (\forall i) \qquad \rho, \vec{e} \Rightarrow \vec{u} \qquad \rho' \cdot \{\vec{f} : \vec{v}\} \cdot \{\vec{x}_j : \vec{u}\}, b_j \Rightarrow r}{\rho, e(e_1, \ldots, e_n) \Rightarrow \text{unwrap}(r)}$$

eval-attr-module
$$\frac{\rho, e \Rightarrow \text{Mod}(q', \rho') \qquad x : v \in \rho'}{\rho, e.x \Rightarrow v}$$

eval-attr-object
$$\frac{\rho, e \Rightarrow \text{Obj}_\Delta(C, \rho')}{\rho, e.x \Rightarrow \rho'(x)}$$

eval-constr
$$\frac{\rho, e_i \Rightarrow v_i \quad (\forall i) \qquad \text{resolve-class}(\rho, q) = C \qquad \text{fields}(C) = x_1, \ldots, x_n, x_{n+1}, \ldots, x_{n+m}}{\rho, q(e_1, \ldots, e_n, y_1{=}e_{n+1}, \ldots, y_m{=}e_{n+m}) \Rightarrow \text{Obj}_\Gamma(C, \{x_1 : v_1, \ldots, x_n : v_n, y_1 : v_{n+1}, \ldots, y_m : v_{n+m}\})}$$

Fig. 20. Expression e evaluates to value v under environment $\rho : \Gamma$

6 Extensions

Beyond the core, PurePy is designed to grow through extensions that add expressiveness while preserving purity. This section sketches two, each drawn from Fluid: partial application and piecewise function definitions. We give an example of each. [**TODO:** Formally define or defer to future work]

6.1 Partial application

A function applied to fewer arguments than it declares returns a function of the remaining arguments:

```
def f(x, y):
    return x + y
```

$\rho, p, v \rightsquigarrow \rho'$

pat-lit

$$\frac{}{\rho, \ell, \ell \rightsquigarrow \varnothing}$$

pat-neg-lit

$$\frac{}{\rho, -n, -n \rightsquigarrow \varnothing}$$

pat-var

$$\frac{}{\rho, x, v \rightsquigarrow \{x{:}\, v\}}$$

pat-wild

$$\frac{}{\rho, _, v \rightsquigarrow \varnothing}$$

pat-list

$$\frac{\rho, p_i, v_i \rightsquigarrow \rho_i \quad (\forall i)}{\rho, [p_1, \ldots, p_n], [v_1, \ldots, v_n] \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_n}$$

pat-tuple

$$\frac{\rho, p_i, v_i \rightsquigarrow \rho_i \quad (\forall i)}{\rho, (p_1, \ldots, p_n), (v_1, \ldots, v_n) \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_n}$$

pat-as

$$\frac{\rho, p, v \rightsquigarrow \rho'}{\rho, p \texttt{ as } x, v \rightsquigarrow \rho' \uplus \{x{:}\, v\}}$$

pat-constr

$$\frac{\begin{array}{c}\text{resolve-class}(\rho, q) = C \qquad C \in \text{ancestors}(C') \qquad \text{fields}(C) = x_1, \ldots, x_n, x_{n+1}, \ldots, x_{n+m} \\ \rho, p_i, \rho''(x_i) \rightsquigarrow \rho_i \quad (\forall i.\, 1 \le i \le n) \qquad \rho, p_{n+j}, \rho''(y_j) \rightsquigarrow \rho_{n+j} \quad (\forall j.\, 1 \le j \le m)\end{array}}{\rho, q(p_1, \ldots, p_n, y_1{=}p_{n+1}, \ldots, y_m{=}p_{n+m}), \text{Obj}_{\Gamma'}(C', \rho'') \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_{n+m}}$$

Fig. 21. Pattern p matches value v under environment ρ, producing bindings ρ'

```
f(5)(6)  # 11
```

The syntax is valid Python, but Python raises a `TypeError` at run time, because `f(5)` is missing an argument. For partial application to add meaning rather than change it, core PurePy must reject such unsaturated calls, requiring every call to supply the declared arity. On that base, `f(5)` denotes the function that adds `5` to its argument, so `f(5)(6)` evaluates to `11`. This is a semantic extension: existing syntax, given a meaning the core withholds.

6.2 Piecewise function definitions

A function may be defined by several clauses that pattern-match on its arguments, in place of a single definition with an explicit `match`:

```
def length(Nil):
    return 0
def length(Cons(x, xs)):
    return 1 + length(xs)
```

Python has no way to define a function by pattern-matching on its arguments: `def length(Cons(x, xs))` is a syntax error, since a parameter cannot be a pattern. Core PurePy, a subset of Python, has none either. The extension adds this: the clauses together define `length`, dispatching on the constructor of the argument. Because Python gives the pattern-clause form no meaning of its own, it is a syntactic extension that needs no special notation to distinguish it from an ordinary `def`.

6.3 Explicit variable definitions

In the core a variable is introduced implicitly by its first assignment, and a definite-assignment analysis (§3) tracks whether it is assigned before it is used. This extension makes the introduction explicit, with a `def` declaration for variables as well as functions:

```
def total = 0   # define and assign
```

```
def n           # define, assignment deferred
if b:
    n = 1
else:
    n = 2
return total + n
```

`def total = 0` declares `total` and assigns it at once, whereas `def n` declares `n` with its assignment deferred, to be supplied later under the existing definite-assignment rules, so `n` is usable only once every path has assigned it. Python has no such form: `def` introduces a function, so a variable `def` is a syntax error, and core PurePy, a subset of Python, has none either. The implicit definitions of the core are the case in which every variable is introduced by `def x = e`, so the core desugars into the explicit form.

7 Case studies

This section reports two research languages for scientific computing that share PurePy's Python-like surface: Fortl and Fluid. [**TODO:** Is there a way to do this without deanonymising the paper?]

7.1 Fortl

Fortl is a statically-typed research language for scientific and numerical computing, with a Python-like syntax. Its distinctive feature is a system of graded numerical types: types are indexed and carry structure, so domain concepts such as units of measure and physical quantities are expressed in the types and enforced by the type checker. A length divided by a time is typed as a speed, and a program that mixes incompatible units is rejected. Fortl aims to let code state the meaning, intent, and limitations of a scientific model, and to make its predictions reproducible and connected to their abstract description. [**TODO:** Confirm and describe the relationship to PurePy]

7.2 Fluid

Fluid is a pure, Pythonic, provenance-aware functional language for transparent research outputs. It is built on PurePy, adding a runtime that records fine-grained dependencies between a computation's inputs and its outputs, so that a figure or value can be linked back to the data it was computed from. Fluid supports pattern matching, destructuring, and partial application, and is aimed at researchers who need both computation and verifiable transparency. As a case study it exercises PurePy as the functional core beneath a substantial additional runtime, and its surface motivates the partial-application and piecewise extensions of §6.

8 Evaluation

We evaluate PurePy empirically along two axes, both concerning how well AI agents work with the subset.

8.1 Generating PurePy

How effectively can AI agents generate PurePy solutions to standard functional-programming problems? This measures how readily the subset can be targeted from a problem statement alone.

8.2 Translating Python to PurePy

How effectively can AI agents translate existing Python programs into the PurePy subset? [**TODO:** Not sure about this one without libraries being part of the spec too.]

9 Related work

9.1 Formal semantics and verified subsets

Although the syntax of mainstream languages is usually described precisely, few define their behaviour formally, so well-defined subsets have been constructed to equip languages with formal, often mechanised, semantics. Examples include Java [Igarashi et al. 2001], OCaml [Seassau et al. 2025], JavaScript [Bodin et al. 2014; Guha et al. 2010; Park et al. 2015; Politz et al. 2012], PHP [Filaretti and Maffeis 2014], Rust [Jung et al. 2020, 2018], and lower-level languages such as WebAssembly [Watt 2021; Watt et al. 2021, 2023] and LLVM-IR [Zakowski et al. 2021; Zhao et al. 2012]. A notable success is Clight [Blazy and Leroy 2009], a substantial subset of C that underpins a verified compiler [Leroy 2009]; likewise the CakeML project [Kumar et al. 2014] produced a verified implementation of a fragment of ML [Milner et al. 1997], building on earlier mechanisations [Lee et al. 2007].

9.2 Formal semantics of Python

Several efforts formalise Python itself: an embedding [Guth 2013] in the $\mathbb{K}$ framework [Roșu and Șerbănuță 2010], a structural operational semantics [Köhl 2021], a reduction of much of the language to a core calculus [Politz et al. 2013], an executable semantics aimed at optimising compilers [Melançon et al. 2023], and a skeletal semantics of a fragment [Andrieux and Schmitt [n. d.]]. PurePy differs in restricting the language to a well-behaved subset rather than modelling Python in full; relating our semantics to these is future work.

9.3 Language subsets for a purpose

A separate line of work delineates subsets for a particular use. The *Source* family [Anderson et al. 2021] comprises progressively larger subsets of JavaScript for teaching first-year students the foundations of computing, following a JavaScript edition of a classic text [Abelson et al. 2022]; the choice of JavaScript was pragmatic, since industrial relevance weighs heavily in language choice for introductory courses [Simon et al. 2018]. For Python, ChocoPy [Padhye et al. 2019] is a statically-typed subset used as the source language in compiler-construction courses, and Restricted Python (RPython) [Ancona et al. 2007] is a statically typed subset that the PyPy project [Rigo and Pedroni 2006] compiles to common runtimes.

9.4 Functional Python

We are aware of one other functional subset of Python [Sunderraman 2025], intended for teaching; PurePy encloses more of the language, since that work omits control structures such as conditionals. Related efforts sit outside the subset discipline: Coconut[4] is a functional language that compiles to Python, and functional pipelines have been proposed for scientific computation in Python [Zhang et al. 2024], alongside index-free data-processing libraries such as Polars.[5] Although PurePy does not aim to make Python more functional, the community has moved in that direction, from lambda expressions in Python 1.0 to a recent proposal for deeply immutable objects [Johnson et al. 2025]. Typed treatments of effects in functional languages, such as Prism,[6] are relevant to the structured effects PurePy may add in a later release.

10 Conclusions and future work

This section summarises the contributions and discusses future work.

[4] https://coconut-lang.org
[5] https://docs.pola.rs/
[6] https://www.stephendiehl.com/posts/prism/

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A Auxiliary definitions

A.1 Syntactic analyses

Figure 22 gives the principal cases of the analyses that support the well-formedness rules. Here $\mathrm{fv}_{\mathrm{e}}(e)$ and $\mathrm{fv}(b)$ are the free variables of an expression and of a block, defined in the usual way. On the cases not shown, binds and $\mathrm{captures}_{\mathrm{e}}$ recurse structurally, taking the union over immediate subterms, and captures takes the union of $\mathrm{captures}_{\mathrm{e}}$ over its subexpressions and captures over its subblocks. assigns is empty on the statements not shown, is the union over the branch blocks of a conditional, and does not descend into nested function definitions, which have their own scope.

A.2 Pattern subsumption

Figure 23 gives the rules for the subsumption preorder $p \sqsubseteq p'$ used by the pattern well-formedness judgement.

A.3 Module and class metafunctions

Figure 24 collects the metafunctions used by the module and class rules.

$$\begin{aligned}
\text{assigns}(x = e) &= \{x\} \\
\text{assigns}(\texttt{def } f_1(\vec{x}_1) :\ b_1 \cdot \ldots \cdot \texttt{def } f_{n+1}(\vec{x}_{n+1}) :\ b_{n+1}) &= \{f_1, \ldots, f_{n+1}\} \\
\text{assigns}(\texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots\ \texttt{case } p_{n+1}\texttt{: } b_{n+1}) &= \textstyle\bigcup_i \big(\text{binds}(p_i) \cup \text{assigns}(b_i)\big) \\
\text{assigns}(\texttt{@dataclass class } c :\ \phi) &= \{c\} \\
\text{assigns}(\texttt{@dataclass class } c(c') :\ \phi) &= \{c\}
\end{aligned}$$

$$\begin{aligned}
\text{binds}(x) &= \{x\} \\
\text{binds}(p\ \texttt{as}\ x) &= \text{binds}(p) \cup \{x\}
\end{aligned}$$

$$\begin{aligned}
\text{captures}(\texttt{def } f_1(\vec{x}_1) :\ b_1 \cdot \ldots \cdot \texttt{def } f_{n+1}(\vec{x}_{n+1}) :\ b_{n+1}) &= \textstyle\bigcup_i \big(\text{fv}(b_i) \setminus (\{\vec{x}_i\} \cup \text{assigns}(b_i))\big) \setminus \{f_1, \ldots, f_{n+1}\} \\
\text{captures}(\texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots\ \texttt{case } p_{n+1}\texttt{: } b_{n+1}) &= \text{captures}_{\text{e}}(e) \cup \textstyle\bigcup_i \big(\text{captures}(b_i) \setminus \text{binds}(p_i)\big)
\end{aligned}$$

$$\text{captures}_{\text{e}}(\texttt{lambda } \vec{x}\texttt{: } e) = \text{fv}_{\text{e}}(e) \setminus \{\vec{x}\}$$

Fig. 22. Assigned, bound, and captured variables: principal cases

$p \sqsubseteq p'$

sub-lit
$$\frac{}{\ell \sqsubseteq \ell}$$

sub-neg-lit
$$\frac{}{-n \sqsubseteq -n}$$

sub-var
$$\frac{}{p \sqsubseteq x}$$

sub-wild
$$\frac{}{p \sqsubseteq _}$$

sub-list-list
$$\frac{p_i \sqsubseteq q_i \quad (\forall i)}{[p_1, \ldots, p_n] \sqsubseteq [q_1, \ldots, q_n]}$$

sub-tuple-tuple
$$\frac{p_i \sqsubseteq q_i \quad (\forall i)}{(p_1, \ldots, p_n) \sqsubseteq (q_1, \ldots, q_n)}$$

sub-as-l
$$\frac{p \sqsubseteq q}{p\ \texttt{as}\ x \sqsubseteq q}$$

sub-as-r
$$\frac{p \sqsubseteq q}{p \sqsubseteq q\ \texttt{as}\ y}$$

Fig. 23. Pattern p is subsumed by pattern p'

$$\text{submodules}(q) = \{x\text{: Mod}(q.x) \mid q.x \in \text{dom}(\mathcal{M})\}$$

(a) submodules metafunction

$$\begin{aligned}
\text{own-fields}(x_1 :\ \texttt{Any} \ \ldots\ x_{n+1} :\ \texttt{Any}) &= x_1, \ldots, x_{n+1} \\
\text{own-fields}(\texttt{pass}) &= \varepsilon
\end{aligned}$$

$$\begin{aligned}
\text{fields}(\text{Cls}_\Gamma(q.c, \vec{x}, \bot)) &= \vec{x} \\
\text{fields}(\text{Cls}_\Gamma(q.c, \vec{x}, c')) &= \text{fields}(\Gamma(c')) \cdot \vec{x}
\end{aligned}$$

$$\begin{aligned}
\text{ancestors}(\text{Cls}_\Gamma(q.c, \vec{x}, \bot)) &= \{\text{Cls}_\Gamma(q.c, \vec{x}, \bot)\} \\
\text{ancestors}(\text{Cls}_\Gamma(q.c, \vec{x}, c')) &= \{\text{Cls}_\Gamma(q.c, \vec{x}, c')\} \cup \text{ancestors}(\Gamma(c'))
\end{aligned}$$

(b) own-fields, fields, and ancestors metafunctions

Fig. 24. Module and class metafunctions